

# Type 2 uncomputability, (polynomial functors) and automata (!)

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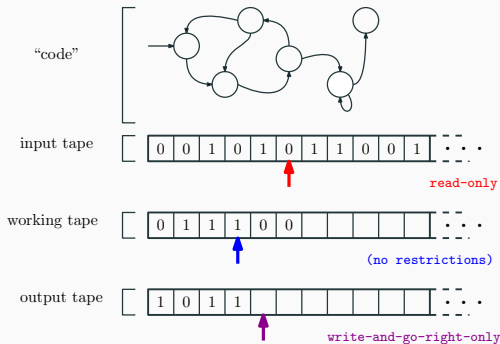
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## $\exists$ type-2 computability

The natural notion of computability for  $2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$

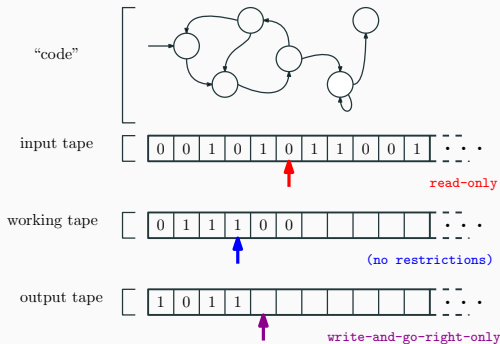


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$$\iff \exists \sqsubseteq\text{-monotone computable } \hat{f} : 2^* \rightarrow 2^* \text{ s.t. } f(p) = \lim_{n \rightarrow +\infty} \hat{f}(p[n])$$

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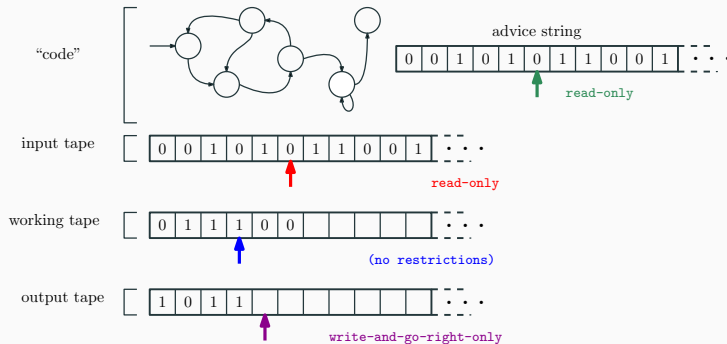
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$$\iff \exists e \in \mathbb{N}. \forall n \in \mathbb{N}. \forall A \in \text{dom}(f). \Phi_e^A(n) = f(A)(n)$$

# Continuity = type-2 computability relativized

Up to a type-1 oracle, all continuous functions are computable.



Formally,  $f : 2^{\mathbb{N}} \rightarrow 2^{\mathbb{N}}$  continuous

$$\iff \exists \sqsubseteq\text{-monotone } \hat{f} : 2^* \rightarrow 2^* \text{ s.t. } f(p) = \lim_{n \rightarrow +\infty} \hat{f}(p[n])$$

$$\iff \exists e \in \mathbb{N}. \exists E \in 2^{\mathbb{N}} \forall n \in \mathbb{N}. \forall A \in \text{dom}(f). \Phi_e^{E \oplus A}(n) = f(A)(n)$$

# Represented spaces

## The category of represented spaces $\mathbf{ReprSp}$

- Objects:  $(X, \delta_X)$  where  $\delta_X$  is a **partial** surjection  $2^\omega \twoheadrightarrow X$
- Morphisms: maps  $X \rightarrow X'$  with a type 2-computable witness

- Super nice: extensive, locally cartesian closed, (co)inductive types

Examples:

- Standard spaces like  $\mathbb{N}$ ,  $2^\mathbb{N}$ ,  $\mathbb{N}^\mathbb{N}$

$$\delta_{2^\mathbb{N}} = \text{id}_{2^\mathbb{N}} \quad \delta_{\mathbb{N}}(0^n 1 p) = n \quad \delta_{\mathbb{N}^\mathbb{N}}(0^{p(0)} 10^{p(1)} 1 \dots) = p$$

- $\mathbb{R}$  via fast Cauchy sequences  $\mathbb{Q}^\mathbb{N}$  (convergence rate  $2^{-n}$ ).
- More generally: all subspaces of (quasi/co)Polish spaces.

# A setting for effective topology (1/2)

We can adapt basic definitions from synthetic topology.

## The Sierpiński space $\mathbb{S}$

As a represented space, define  $\delta_{\mathbb{S}} : 2^{\mathbb{N}} \rightarrow \{\perp, \top\}$  by

$$\delta_{\mathbb{S}}(x) = \perp \quad \Longleftrightarrow \quad x = 0^\omega$$

## The spaces of open and closed subsets of $X$

We set  $\mathcal{O}(X) = \mathcal{A}(X) = \mathbb{S}^X$ , dual interpretations of  $\in$

$$x \in U \in \mathcal{O}(X) \Longleftrightarrow U(x) = \top \quad x \in F \in \mathcal{A}(X) \Longleftrightarrow F(x) = \perp$$

- Code for an open = eventually we notice if we are in
- Computably open =  $\in$  is semi-computable
- $\bigcup : \mathcal{O}(\mathcal{O}(X)) \rightarrow \mathcal{O}(X)$ ,  $\cup$ ,  $\cap$ , computable, ...

## A setting for effective topology (2/2)

### The space of compact subsets $\mathcal{K}(X)$

A subspace  $\mathcal{K}(X) \subseteq \mathcal{O}(\mathcal{O}(X))$ ; concretely  $\delta_{\mathcal{K}(X)}(p) = A \subseteq X$  iff

- we have  $f \in \mathbb{S}^{\mathbb{S}^X}$  with  $\delta_{\mathcal{O}(\mathcal{O}(X))}(p) = f$
- for all open  $U \in \mathcal{O}(X)$ ,  $f$  detects if  $U$  covers  $A$

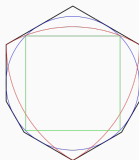
$$A \subseteq U \iff f(U) = \top$$

- $X$  is computably compact iff  $\forall_X : \mathcal{O}(X) \rightarrow \mathbb{S}$  is computable
- $[0, 1]^\omega$ ,  $2^{\mathbb{N}}$  computably compact...
- There exists a dual notion to compactness which is trivial classically, but relevant computably.

# An application: Universal Lebesgue covering Problem

## Theorem (Pauly & Ruggles '23)

The area of the smallest convex shape in  $\mathbb{R}^2$  that cover all shapes of diameter 1 **is a computable real**.



Proof ideas:

- Show separately it is an upper/lower-semicomputable real
- $\inf U$  is upper-semi computable for a computable  $U \in \mathcal{O}(\mathbb{R})$
- Use effective compactness and overtness over various things: compact subsets of  $[0, 1]^2$ ,  $\text{GL}_2([-2, 2])$ ,  $\dots$

# Non-computable, yet interesting tasks

- For given  $x, y \in X$ , do we have  $x = y$  or not?  $(=_X)$
- Given  $p \in 2^{\mathbb{N}}$ , does  $p = 0^\omega$  or not? (LPO)
- Given  $x, y \in \mathbb{R}$ , do we have  $x \leq y$  or  $y \leq x$ ? (LLPO)
- Given a non-empty  $A \in \mathcal{A}(X)$ , find  $x \in A$ .  $(C_X)$
- From  $a, b \in \mathbb{R}$ , find  $x$  such that  $ax = b$ .  $(\text{urbDiv}, \dots)$
- Given a colouring  $[\mathbb{N}]^k \rightarrow n$ , produce an homogeneous set.  $(\text{RT}_n^k)$
- Find paths in various classes of  $\omega$ -branching trees.  $(\text{WKL}_0, \dots)$
- ...

## Motivation for Weihrauch complexity

Compare the relative hardness of these tasks.

# Weihrauch problems

## Partial Weihrauch problems, formally

Families  $(P_i)_{i \in \text{dom}(P)}$  with  $\text{dom}(P) \subseteq 2^{\mathbb{N}}$  and  $P_i \subseteq 2^{\mathbb{N}}$  for every  $i$ .

( $\simeq$  internal families in regular projective represented spaces)

- $\text{dom}(P)$  = set of **questions**
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Examples:

- LPO: “Given  $p \in 2^{\mathbb{N}}$ , return 0 iff  $p = 0^\omega$ ”:

$$\text{LPO}_{0^\omega} = \{0^\omega\}$$

$$\text{LPO}_p = \{10^\omega\}$$

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- $\text{C}_{\mathbb{N}}$ , up to messing with representations for  $\mathbb{N}$  and  $\mathbb{N}^{\mathbb{N}}$ :

$$\begin{aligned} \text{dom}(\text{C}_{\mathbb{N}}) &= \{p \in \mathbb{N}^{\mathbb{N}} \mid \exists n \in \mathbb{N}. n \notin \text{range}(p)\} \\ (\text{C}_{\mathbb{N}})_p &= \{n \in \mathbb{N} \mid n \notin \text{range}(p)\} \end{aligned}$$

# Weihrauch reducibility

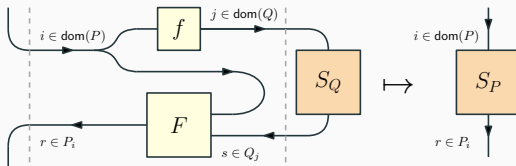
TL;DR: almost like Turing reducibility, but

- adapted to type 2 computability
- reductions must make **exactly** one oracle call
- (and Weihrauch problems are “non-deterministic”)

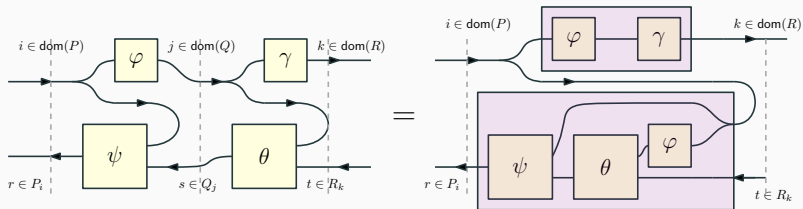
## Official definition

$P \leq_W Q$  if there are **computable**

$$f : \text{dom}(P) \rightarrow \text{dom}(Q) \quad \text{and} \quad F : \prod_{i \in \text{dom}(P)} (Q_{f(i)} \rightarrow P_i)$$

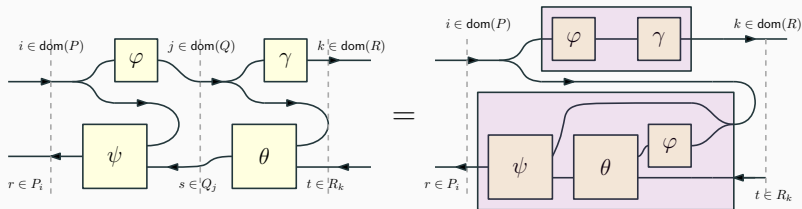


# Sanity checks



Reductions compose + Quotienting by  $\equiv_W \rightsquigarrow$  **Weihrauch degrees**

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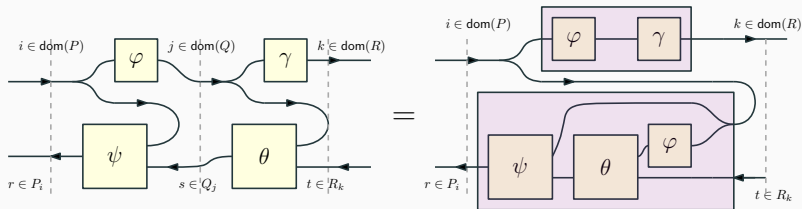
**Abstractly: reductions are container morphisms**

Weihrauch problems = containers over represented subspaces of  $2^{\mathbb{N}}$

Makes sense in over any category with pullbacks

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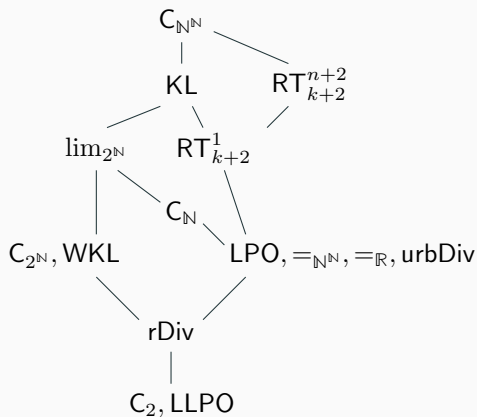
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For LCCCs: “reductions”  $\cong$  strong natural transformations between polynomial functors  $X \mapsto \sum_{i \in \text{dom}(P)} X^{P_i}$

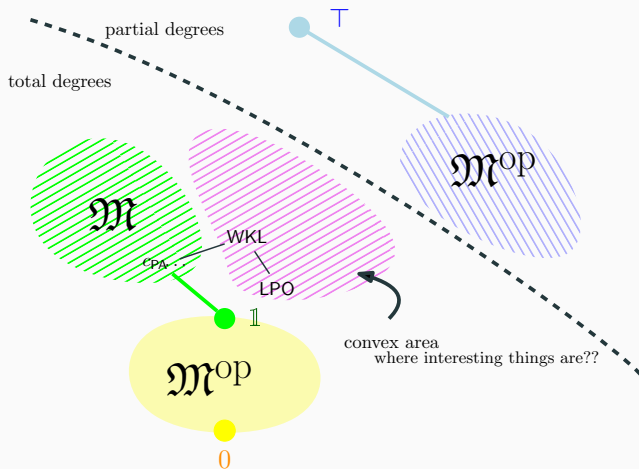
## How some problems we've seen so far relate



One can certainly draw more ambitious zoos...

# Zooming out

The (partial) Weihrauch lattice is big:  $2^{2^{\aleph_0}}$ .

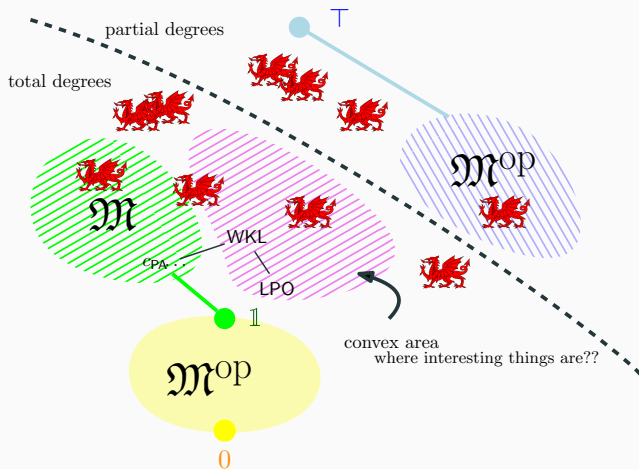


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# Technical asides: encoding and intensionality

**Legitimate question: why subsets of  $\mathbb{N}^{\mathbb{N}}$ ?**

Why not take internal families/containers in *represented spaces*?

- It would make sense and the category is nicer
- But computability theorists want reductions to be **intensional**, which yields more identifications than working extensional.

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- Containers over a **weakly** locally cartesian closed category
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(or developing the theory of polynomial quasifunctors if you're volunteering)

# Algebras of problems

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# Some algebra on problems

A lot of generic nice structure.

- Joins  $+$ : “solve one of the two problems”

$$\begin{array}{lcl} \text{dom}(P + Q) \cong \text{dom}(P) + \text{dom}(Q) & (P + Q)_{\text{in}_1(i)} & = P_i \\ & (P + Q)_{\text{in}_2(j)} & = Q_j \end{array}$$

- Meets  $\wedge$ : “given inputs for both, solve one”

$$\text{dom}(P \wedge Q) \cong \text{dom}(P) \times \text{dom}(Q) \quad (P \wedge Q)_{i,j} = P_i + Q_j$$

- Parallel product  $\otimes$ : “solve both problems”

$$\text{dom}(P \otimes Q) \cong \text{dom}(P) \times \text{dom}(Q) \quad (P \otimes Q)_{i,j} = P_i \times Q_j$$

Exercise: what are the units? What are the corresponding operations on polynomials?

## Some examples

- Most relationships with  $+$ ,  $\wedge$   
 $\leadsto$  deduce from the theory of distributive lattices.
- $\text{LPO} <_{\text{W}} \text{LPO} \otimes \text{LPO} <_{\text{W}} \dots <_{\text{W}} \text{LPO}^n$
- $\text{LPO}^n <_{\text{W}} \text{RT}_2^2$
- Some problems  $P$  are **finitely parallelizable**:

$$P \otimes P \leq_{\text{W}} P \quad \text{and} \quad \mathbb{1} \leq_{\text{W}} P$$

Examples:  $\text{WKL}_0$ ,  $\lim_{2^{\mathbb{N}}}$ ,  $\text{C}_{\mathbb{N}}$ ,  $\text{KL}$ ,  $\text{C}_{\mathbb{N}^{\mathbb{N}}}$

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### Definition (?)

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$$\text{dom}(P^*) = \text{dom}(P)^* \qquad P_w = \prod_{i \in |w|} P_{w_i}$$

# One oracle call after the other

## Sequential composition $P \star Q$

- Implicitly: ability to make an oracle call to  $Q$  then  $P$
- Explicitly: given an instance  $i$  of  $Q$  and a function that takes a solution of  $i$  to an instance of  $P$ , compute all relevant solutions

$$\begin{aligned}\text{dom}(P \star Q) &= \sum_{i \in \text{dom}(Q)} (Q_i \rightsquigarrow \text{dom}(P)) \\ (P \star Q)_{i,f} &= \sum_{r \in Q_i} P_{f(r)} \quad \text{(using the weak exponential } \rightsquigarrow \dots \text{)}\end{aligned}$$

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## Extensional version, on polynomials

$$\left( \sum_{i \in \text{dom}(P)} X^{P_i} \right) \circ \left( \sum_{j \in \text{dom}(Q)} X^{Q_j} \right) \cong \sum_{\substack{j \in \text{dom}(Q) \\ f: Q_j \rightarrow \text{dom}(P)}} X^{\sum_r P_{f(r)}}$$

## Some examples

- Obviously,  $P \otimes Q \leq_W P \star Q$ .
- $\text{LPO} \otimes \text{LPO} <_W \text{LPO} \star \text{LPO} <_W \text{LPO} \otimes \text{LPO} \otimes \text{LPO}$
- Some problems  $P$  are closed under sequential composition

$$P \star P \leq P \qquad 1 \leq P$$

Examples:  $\text{C}_{\mathbb{N}}$ ,  $\text{WKL}_0$ ,  $\text{C}_{\mathbb{N}^{\mathbb{N}}}$

### Iterated composition $P^{\diamond}$

- Explicitly: computed as the least fixpoint of  $X \mapsto 1 + X \star P$
- Implicitly: ability to make a finite but not fixed in advance number of oracle calls to  $P$

# Automata link 1: Fixpoints of problems

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# Motivation

We've seen some useful **least fixpoints**.

## Other useful fixpoints

**Infinite parallelization**  $\hat{P}$ :      fixpoint of  $X \mapsto P \otimes X$

$$\text{dom}(\hat{P}) = \text{dom}(P)^{\mathbb{N}} \qquad \hat{P}_s = \prod_{n \in \mathbb{N}} P_{s_n}$$

**Infinite sequential iteration**  $P^\omega$ :      fixpoint of  $X \mapsto X \star P$

Construction as an inverse limit left to the listener

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Exercise: these are **not** greatest fixpoints!

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# Fixpoint of operators

## Existence of useful fixpoints (P. & Price '26)

If  $F$  is a nice enough endofunctor over containers, over a nice enough category (whatever that means), the following exists:

- an initial algebra  $\mu F$  for  $F$
  - a terminal coalgebra  $\nu F$  for  $F$
  - a somewhat canonical (co)algebra  $\zeta F$  sitting in-between
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- Idea: construct a fixpoint on domains, and then on the answers.
  - Application:  $\hat{P} \equiv_{\mathbf{W}} \zeta X. X \otimes P$  and  $P^\omega$  morally like  $\zeta X. X \star P$ .

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## Meta-question (I don't know the answer)

Is this relevant for other applications of containers?

# Towards an automata-theoretic problem

We were drawn to studying (total parts) of problems generated by the following closed  $\zeta$ -expressions:

$$P, Q ::= X \mid \gamma X.P \mid P \square Q \quad \square \in \{+, \otimes, \wedge\}, \gamma \in \{\mu, \nu, \zeta\}$$

They're all generated by regular infinite **game-tree** languages.

## Game-tree problems

Given an infinite regular tree language

$$L \subseteq \text{Tree}(\{\exists, \forall\} \times \{0, \dots, k\}) \quad (\text{a tree} = \text{an arena for a parity game})$$

define the problem  $\llbracket L \rrbracket$  by  $\text{dom}(\llbracket L \rrbracket) = L$  and

$$\llbracket L \rrbracket_G = \text{winning strategies for Eve in } G$$

Most problems we've seen today are equivalent to game-tree problems.

# A purely automata-theoretic question?

## Conjecture

From automata recognizing  $L, L'$ , we can decide “ $\llbracket L \rrbracket \leq_W \llbracket L' \rrbracket$ ?”.

- Very easy exercise: with  $\llbracket L \rrbracket \equiv_W 1$
- Maybe exercise (???): with  $\llbracket L' \rrbracket \equiv_W 1$ 
  - $\llbracket L \rrbracket \leq_W 1$  close to “one can continuously turn game trees in  $L$  to a winning strategy for the relevant player”.
- Not sure about the necessary technology for the more general case

# A purely automata-theoretic question?

## Conjecture

From automata recognizing  $L, L'$ , we can decide “ $\llbracket L \rrbracket \leq_W \llbracket L' \rrbracket$ ?”.

- Very easy exercise: with  $\llbracket L \rrbracket \equiv_W 1$
- Maybe exercise (???): with  $\llbracket L' \rrbracket \equiv_W 1$ 
  - $\llbracket L \rrbracket \leq_W 1$  close to “one can continuously turn game trees in  $L$  to a winning strategy for the relevant player”.
- Not sure about the necessary technology for the more general case
- ( $\exists$  a variant of the conjecture in trees with coloured holes in the paper)

## Automata link 2: Variants of Kleene algebras

---

# Equational theory of the Weihrauch lattice

- The Weihrauch degrees are a distributive lattice.
- Every countable distributive lattice embeds into  $(\mathfrak{W}, +, \wedge)$   
(via the Medvedev degrees)
- Thus,  $(\mathfrak{W}, +, \wedge) \models t \leq u$  iff  $t \leq u$  is provable from the axioms of distributive lattices. (formulas being implicitly universally quantified)

## Driving question

Can we extend this to additional operations? In particular:

- Can we axiomatize equation in those extensions?
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## Meta-question

For a given signature, is there anything true in the Weihrauch degree that is not true for **all** (suitable) categories of containers?

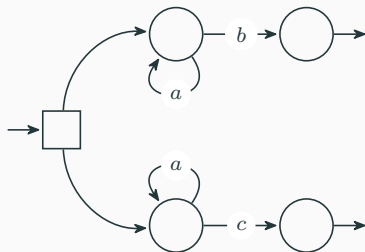
# Terms with composition and automata

## Starting observation

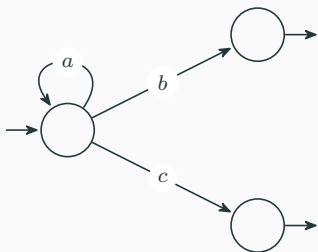
Terms over  $0, 1, +, \star, (-)^\diamond, \top$  = almost regular expressions.

(alphabet = the set of variables)

- Terms can be mapped to NFAs in a meaningful way
- Adding  $\wedge$  = allowing alternating automata



$b \star a^\diamond \wedge c \star a^\diamond$



$(b + c) \star a^\diamond$

# Universal validity and games

Given alternating automata  $\mathcal{A}$  and  $\mathcal{B}$ , we can define a game  $\mathcal{D}(\mathcal{A}, \mathcal{B})$  that captures a variant of the notion of **simulation** such that

## Theorem

$(\mathfrak{W}, \mathbb{1}, +, \top, \wedge, \star, (-)^\diamond) \models t \leq u$  iff Duplicator wins in  $\mathcal{D}(\mathcal{A}_t, \mathcal{A}_u)$ .

Some properties of  $\mathcal{D}(\mathcal{A}, \mathcal{B})$ :

- this is a Büchi game
- allows to make several attempts at simulating  $\mathcal{A}$  in parallel  
(using  $\mathcal{B}$  exactly once)

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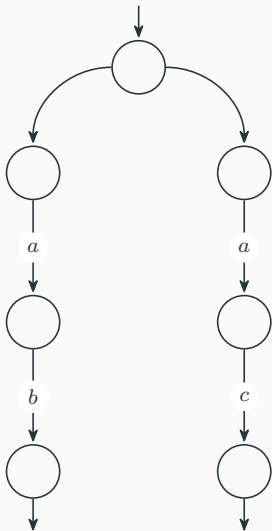
$\Rightarrow \mathcal{O}(|\mathcal{B}|2^{|\mathcal{A}|})$  positions

## Corollary

“ $(\mathfrak{W}, \mathbb{1}, 0, \top, +, \wedge, \star, (-)^\diamond) \models t \leq u$ ?” is decidable.

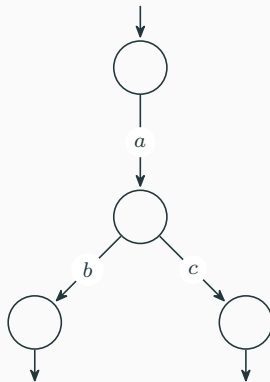
- Conjecture: this is PSPACE-complete.

# A simple example of simulation and non-simulation



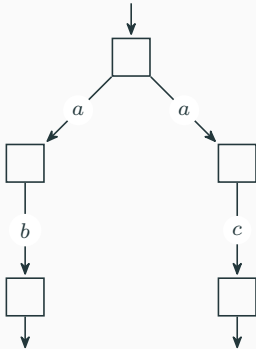
$(b \star a) + (c \star a)$

$<$



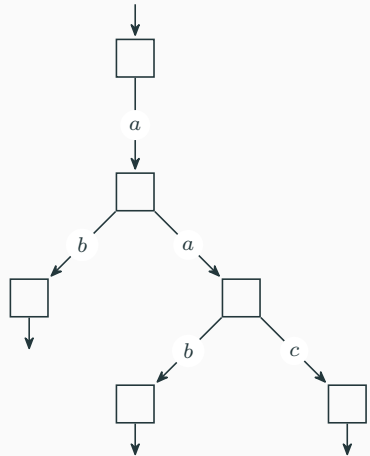
$(b + c) \star a$

# A simulation requiring several concurrent attempts



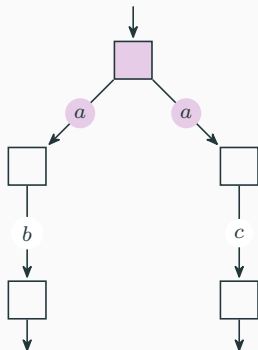
$$(b \star a) \wedge (c \star a)$$

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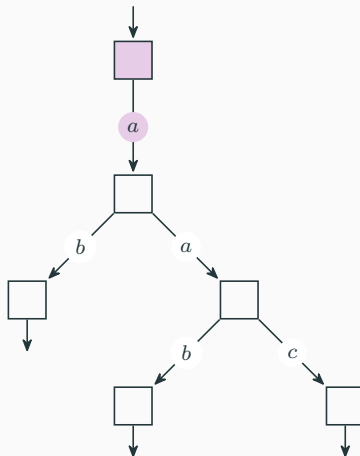
$$(((c \wedge b) \star a) \wedge b) \star a$$

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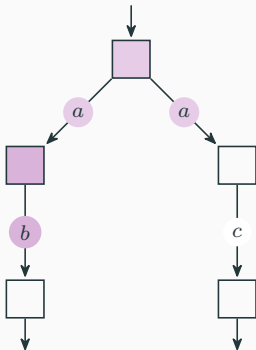
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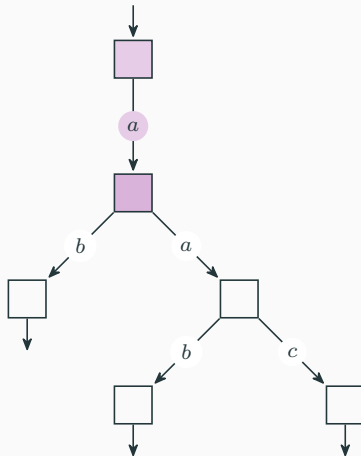
Let's play ?!

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$$(b \star a) \wedge (c \star a)$$

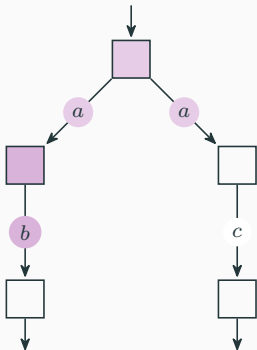
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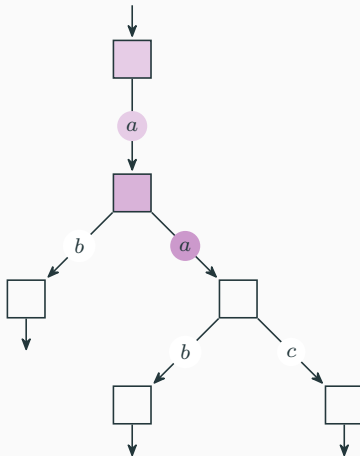
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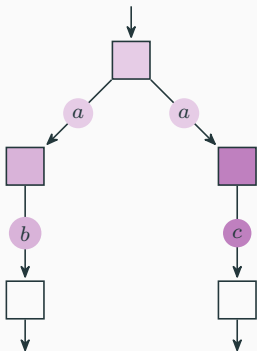


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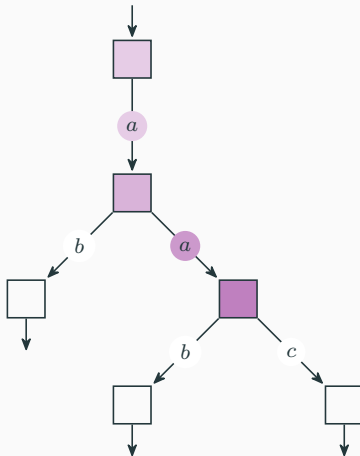
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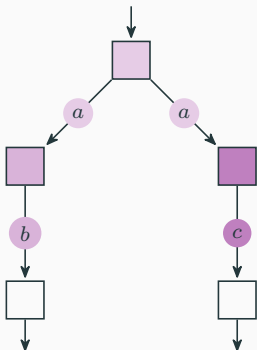
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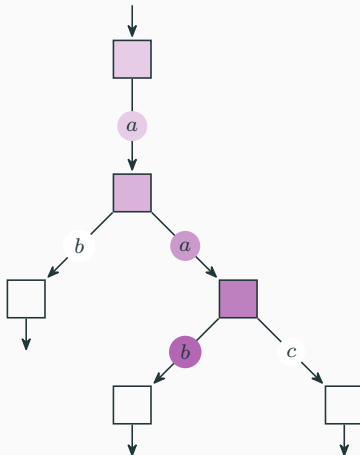
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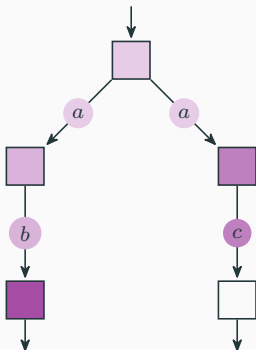


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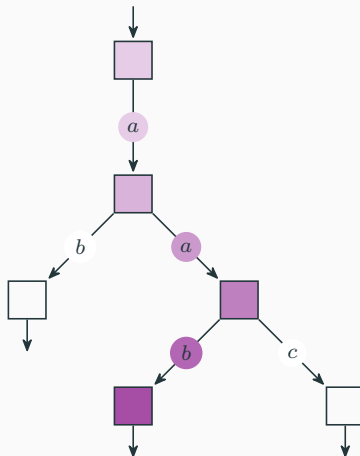
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Let's play ?!

# A simulation requiring several concurrent attempts



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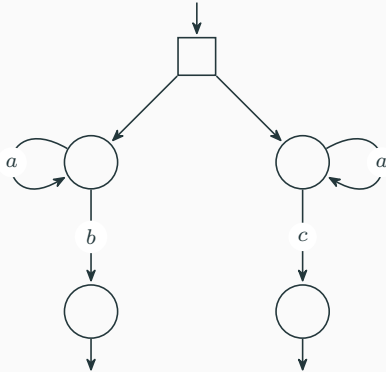


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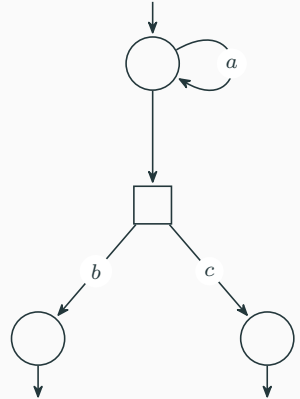
Let's play ?!

## Another simulation requiring several concurrent attempts



$$(b \star a^\diamond) \wedge (c \star a^\diamond)$$

$\equiv$



$$(b \wedge c) \star a^\diamond$$

## Where do the concurrent attempt come from

- $\text{dom}(a \wedge b) = \text{dom}(a) \times \text{dom}(b)$ ,
- Solving either question is progress along a possible run
- Conceptually captured in the following **strength** axiom:

$$a \star b \wedge c \leq_{\text{w}} (a \wedge c) \star b$$

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### For category theorists

Corresponds to the canonical strength in polynomial functors.

$$P(X) \times Y \longleftarrow P(X \times Y)$$

# Induction principles for $(-)^{\diamond}$

## Non-trivial useful axiom for fixpoints (Westrick, 2021)

The following is valid in the Weihrauch degrees

$$x \star x \leq x \quad \Rightarrow \quad x^{\diamond} \leq x$$

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(exercise:  $\mathbb{1} \leq a \wedge b$  implies  $a^{\diamond} \wedge b^{\diamond} \leq (a \wedge b)^{\diamond}$ )

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## Theorem

The above axioms are valid in the partial Weihrauch degrees.

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Candidate axiomatization of inequations

- All the axioms of LKA **w/o right-distributivity of  $+$  over  $\star$**

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Complete for the equational theory of  $(\mathfrak{W}, \mathbb{1}, 0, \top, +, \wedge, \star, (-)^\diamond)$ .

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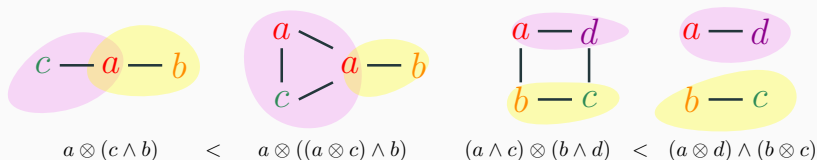
Complete for the equational theory of  $(\mathfrak{W}, \mathbb{1}, 0, \top, +, \wedge, \star, (-)^\diamond)$ .

Proof idea:  $\exists$  positional simulation strategies, induction on the syntax

# How this started: the theory of $\otimes, \wedge$

## A notion of combinatorial reduction between graphs

A reduction from  $(V_0, E_0, c_0)$  to  $(V_1, E_1, c_1)$  is a colour-preserving function  $h : V_1 \rightarrow V_0$  such that the image of any maximal clique in  $(V_1, E_1)$  under  $h$  contains a maximal clique in  $(V_0, E_0)$ .



## Combinatorial characterization (Neumann, Pauly, P.)

$(\mathfrak{W}, \wedge, \otimes) \models t \leq u$  iff there is a reduction from  $G_t$  to  $G_u$ .

As a result, deciding  $(\mathfrak{W}, \wedge, \otimes) \models t \leq u$  is  $\Sigma_2^P$ -complete.

- Axiomatizing: harder!
- $+$ ,  $\star$  and  $\otimes$ : opens the gates of hell (**concurrency theory**)

# Concluding thoughts on this stuff

## I feel ignorant (help?)

- Is this relevant/known to non-computability theorists?
- Do people care about the algebra of strong monotone operators on complete Heyting algebras?
- Literature on simulation between alternating automata?

## Avenues for further work

- Horn theory?
- Fun extensions to the signature:
  - Easyish:  $\mu$ -calculus like-formalism with  $\star, \wedge, \vee$
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Thanks for listening! Questions?